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SECJ 3303 – INTERNET PROGRAMMING

TOPIC 1 – AN INTRODUCTION TO WEB PROGRAMMING WITH JAVA

innovative • entrepreneurial • global



UTM JOHOR BAHRU

OVERVIEW



SYNOPSIS

This course exposes the students to the current methods for analyzing, designing, developing, and deploying the web applications using Java technology.

This course covers the development of web component with Servlet, Java Server Pages (JSP) and Java Web Spring Framework.

The students should obtain the knowledge and skills necessary to quickly build web applications based on Servlet and JSP technologies using the STS (Sprint Tool Suite 4) IDE, Apache Maven tool to build and manage dependencies, Tomcat web container and Java Spring Framework.

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COURSE LEARNING OUTCOMES

CLO 1  Apply the concept of static and dynamic web applications development.

Develop dynamic web applications based on J2EE™ platform that using technology such as Servlet, JSP and JavaBeans.  **CLO 2**

CLO 3  Construct web application using appropriate software methods and tools

Work in a team to develop a medium to complex dynamic web application as a group mini project using J2EE platform.  **CLO 4**

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TOPICS

01 INTRODUCTION TO WEB DEVELOPMENT

02 SERVLET PROGRAMMING

03 JSP PROGRAMMING

04 JAVABEAN

05 MVC ARCHITECTURE

06 SESSION MANAGEMENT

07 SPRING MVC

08 DATABASE

09 EXPRESSION LANGUAGE (EL) & JSP STANDARD TAGS LIBRARY (JSTL)

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OBJECTIVES

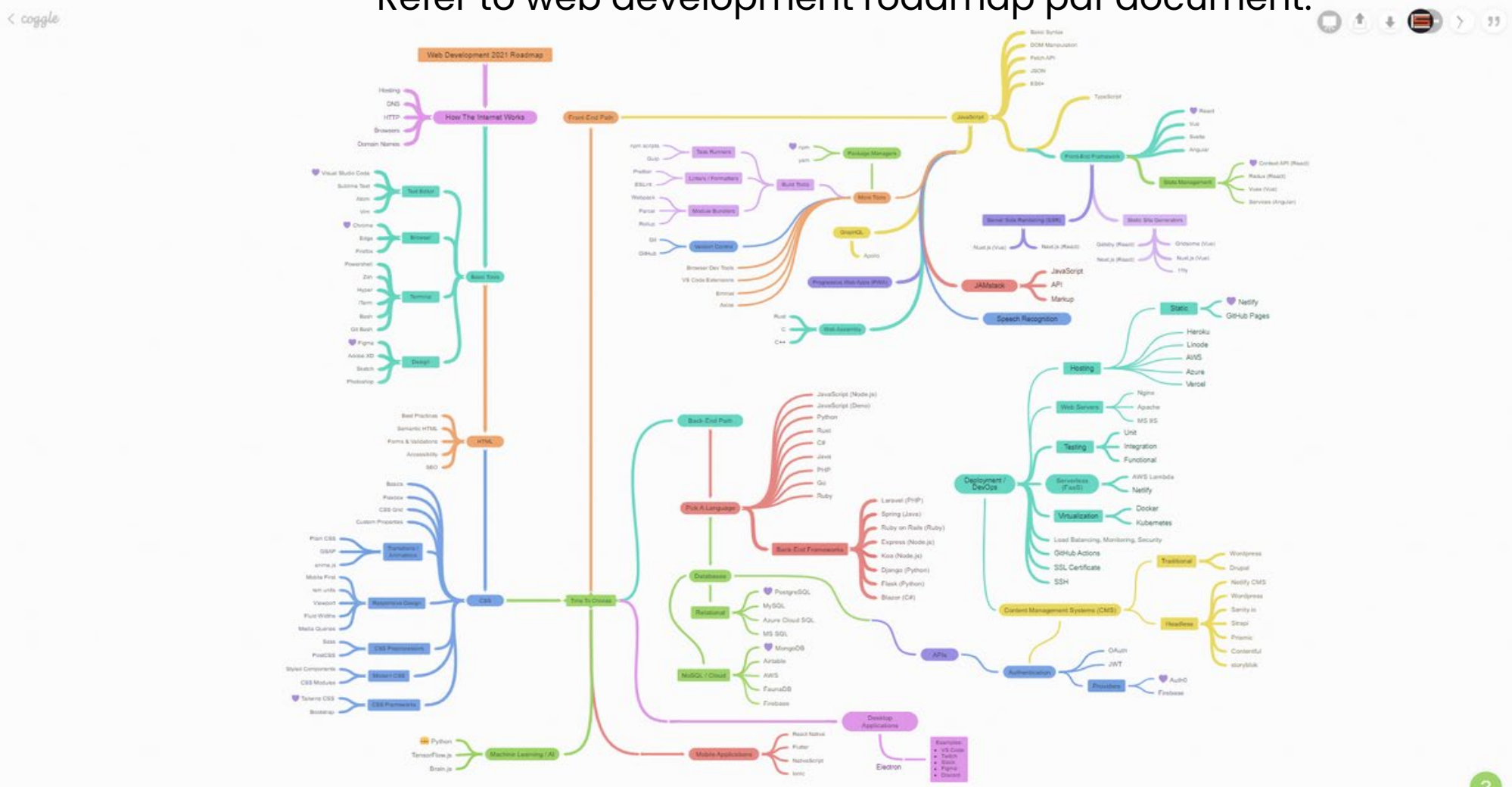
- Understand web development roadmap
- Discuss concept of web application and its components
- Distinguish between HTML and HTTP
- Distinguish between static and dynamic web pages
- Identify approaches for developing web application
- Understand the components and architecture of servlet/JSP application
- Identify the appropriate tools/technologies for developing web application

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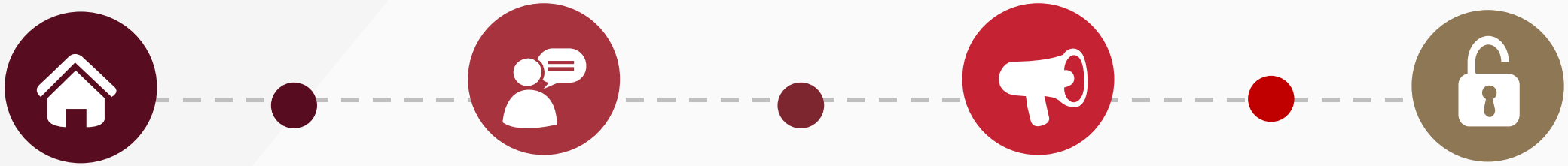
OVERVIEW

WEB DEVELOPMENT ROADMAP

**Refer to web development roadmap pdf document.



BASIC WEB DEVELOPMENT ROADMAP



What is web application?

- How web application works?
- Front-End and Back-End
- Code Editor

Basic Front-End

- HTML
- CSS
- Javascript

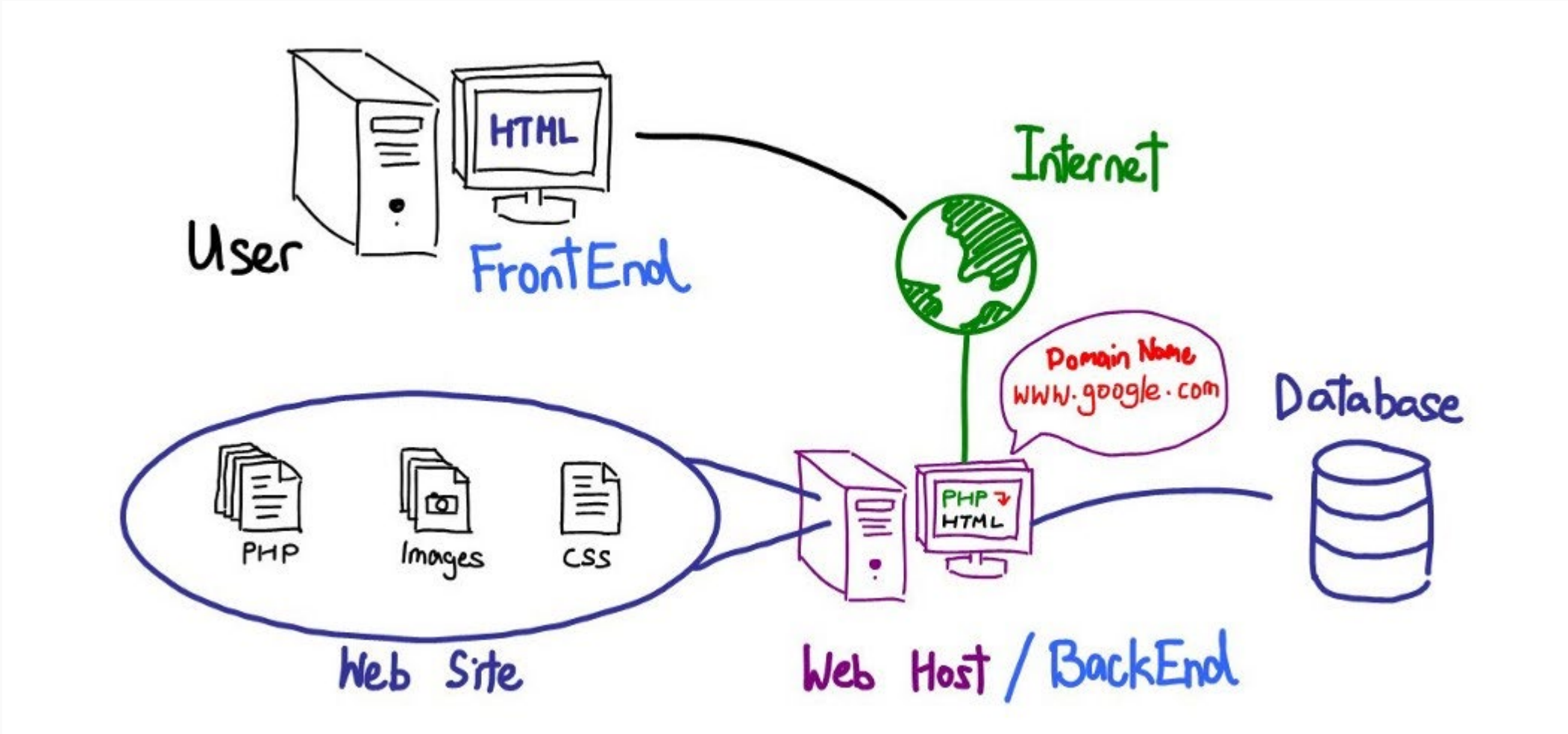
Tools

- Package Manager
- Build Tool
- Version Control

Basic Back-End

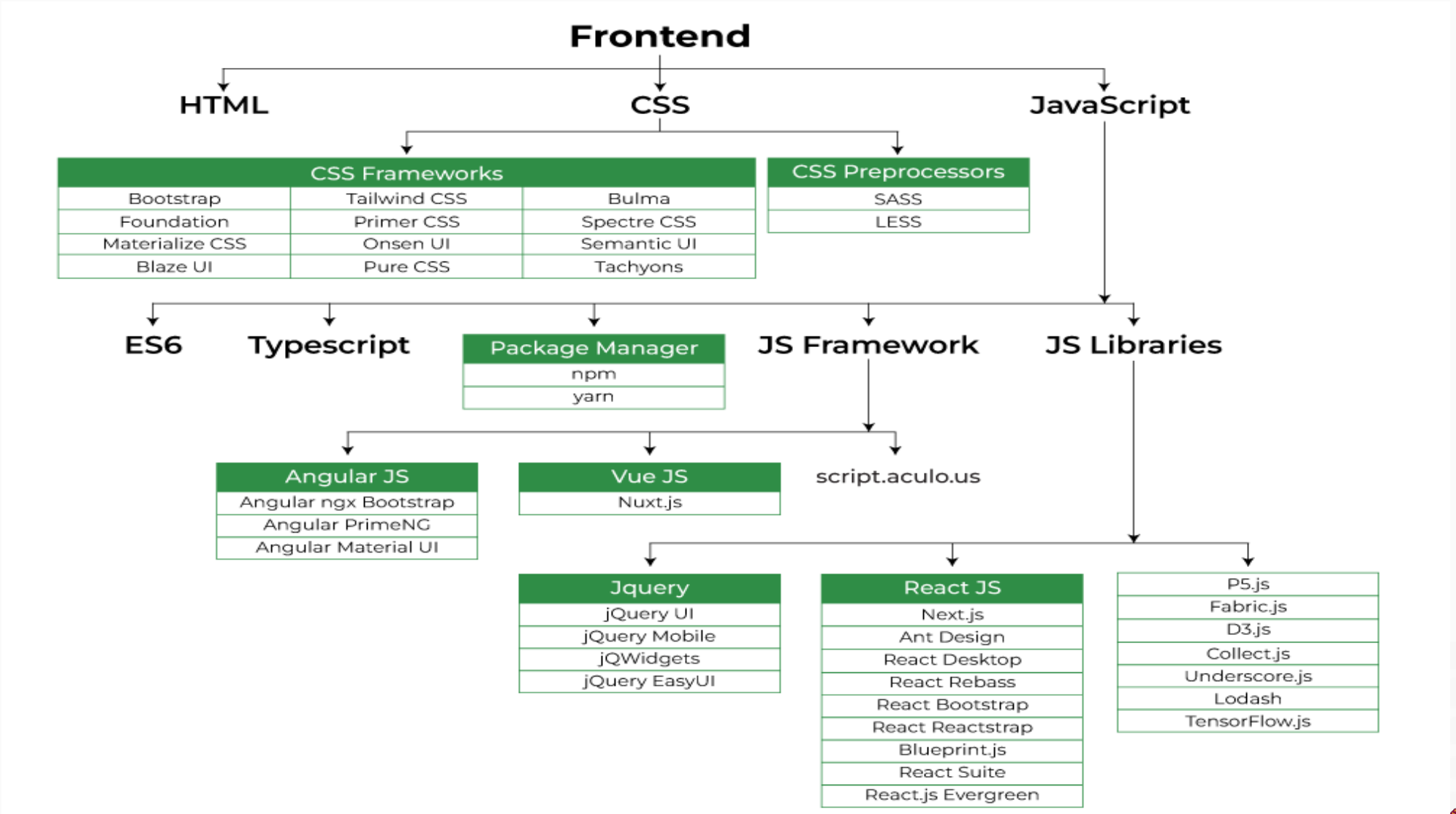
- Server
- Programming Language
- Database

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HTML
User
FrontEnd
Internet
Domain Name
www.google.com
Database
PHP
HTML
PHP
Images
CSS
Web Site
Web Host / BackEnd

TOPIC 1

FRONT-END ROADMAP



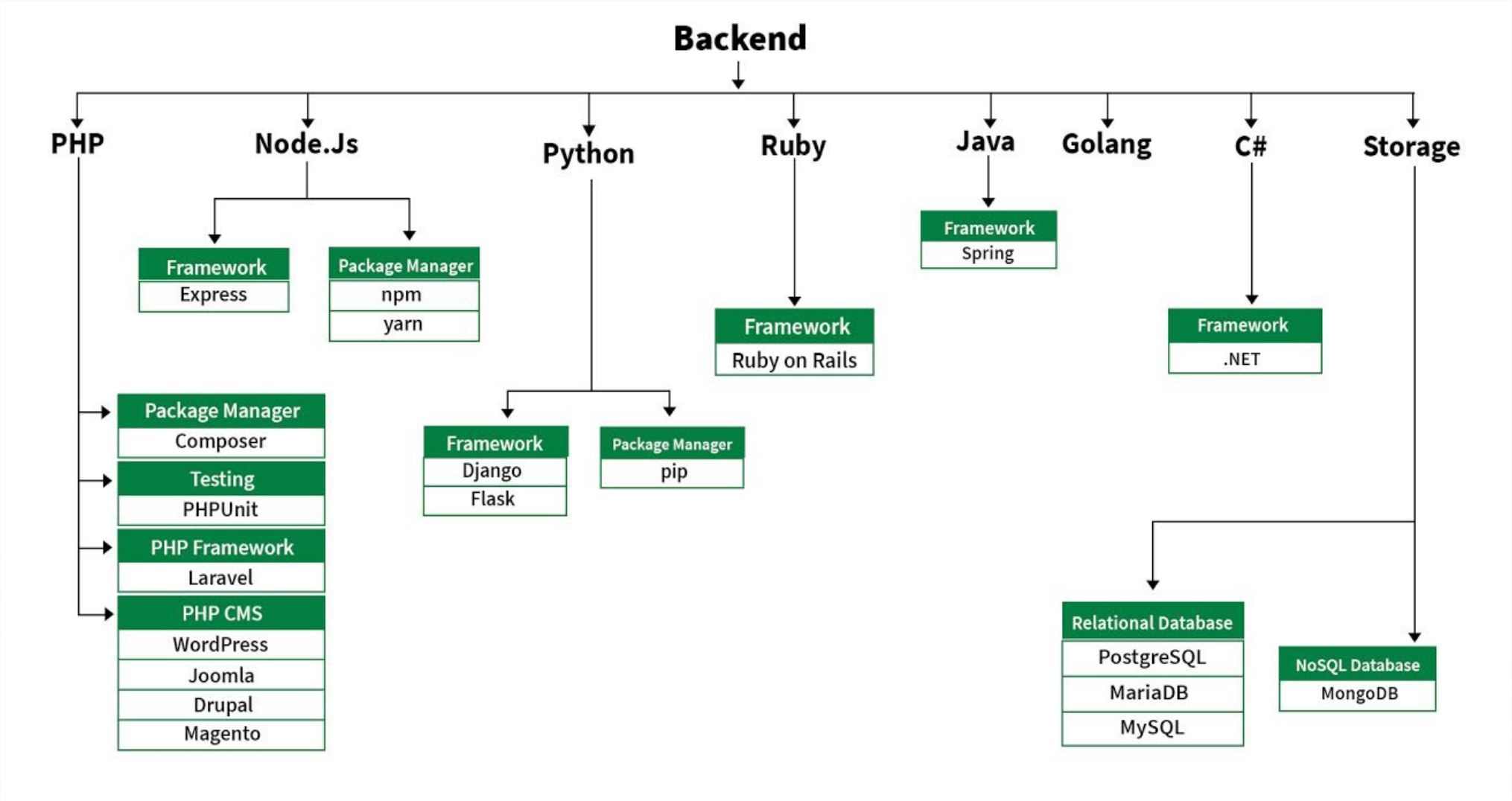
TOOLS

- **Package Manager:** Example are NPM and Yarn.
- **Build Tool:** Programs that automate the process of building an executable application from source code. This building process includes activities like compiling, linking and packaging the code into an executable form. Example of build tool are Sublime Text and Chrome Developer Tool.
- **Version Control:** When certain libraries are added to each application, it gets updated in a period which in turn creates a new version of it. Example of version control are GitHub, GitLab, Beanstalk, Apache Subversion, Mercurial.
- **Benefits of Version Control:**
    - It helps to maintain code.
    - Can re-use it whenever needed.
    - Saves the changes made.
    - Provides evidence of all changes made.
    - Improves performance of the web development.
    - Collaborate and contribute.

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BACKEND ROADMAP



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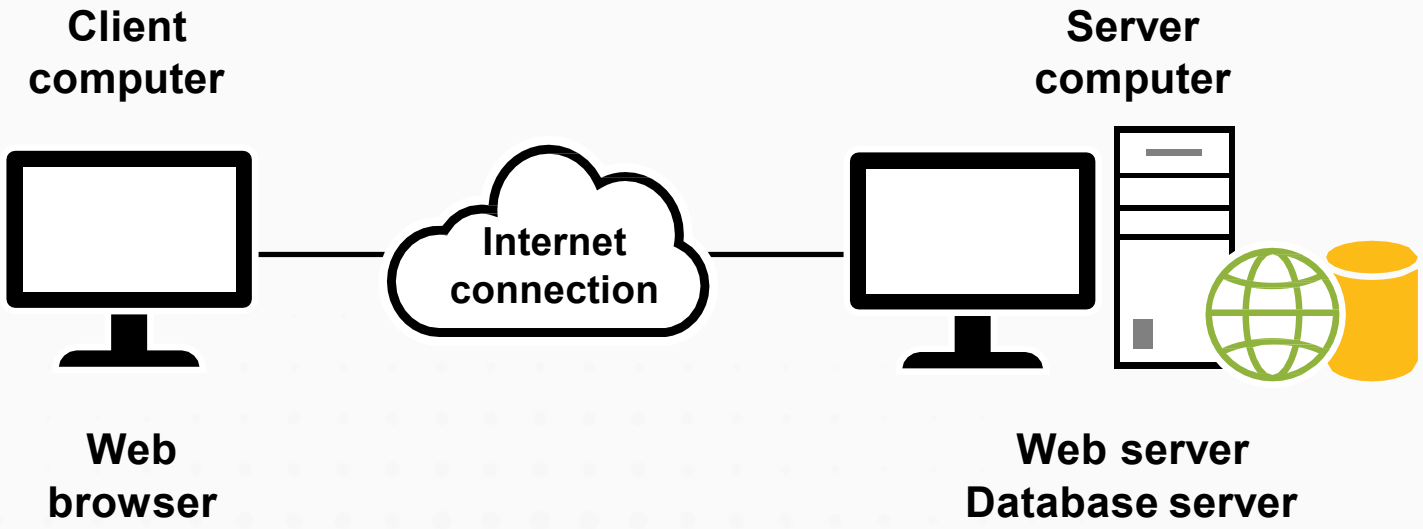
WEB APPLICATION

- A set of web pages that are generated in response to user request.
- A type of client-server application – a user at a client computer accesses an application at a server computer connected via the Internet or an intranet.
- The user works with a web browser at the client computer. The web browser provides the user interface for the application. Example of web browsers: Google Chrome, Mozilla Firefox, Safari etc.
- Runs on the server computer under the control of web server software. The Apache is one of the most widely used web server.
- The server computer also runs a Database Management System (DBMS), which is also known as a database server.
- For Servlet and JSP applications, Oracle and MySQL are the most popular DBMS.

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TOPIC 1

COMPONENTS OF WEB APPLICATION



STATIC WEB PAGES

TOPIC 1

- Hypertext Markup Language (HTML) is the language that the web browser converts into the web pages of a web application.
- A static web page is an HTML document that is stored in a file and does not change in response to user input. Filename with an extension of *.htm* or *.html*.
- Hypertext Transfer Protocol (HTTP) is the protocol that the web browsers and web servers use to communicate.
- A web browser requests a page from a web server by sending the server a message known as HTTP request. The HTTP request includes the name of the HTML file that is requested.
- A web server replies to an HTTP request by sending a message known as HTTP response back to the browser. The HTTP response includes the HTML document that is stored in HTML file.

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DYNAMIC WEB PAGES

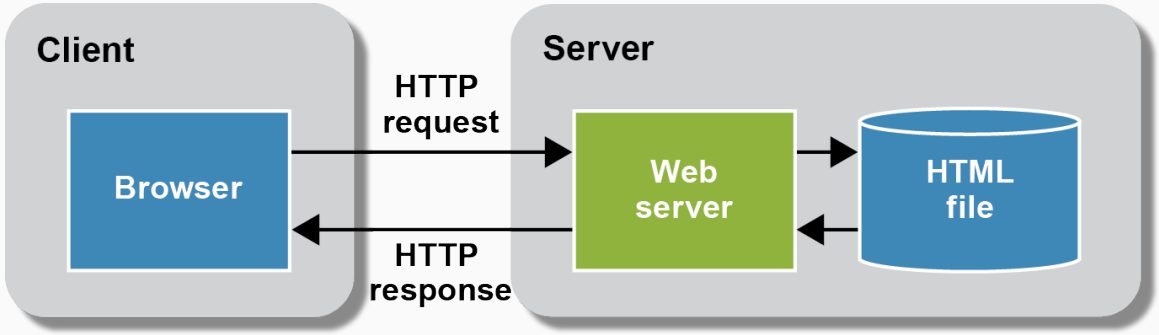
TOPIC 1

- A dynamic web page is an HTML document that is generated by a web application. Often, the web page changes according to parameters that are sent to the web application by the web browser.
- How it works?
  - ❑ When a web server receives a request for a dynamic web page, the server passes the request to the web application.
  - ❑ Then, the application generates a response, such as an HTML document, and returns it to the web server.
  - ❑ The web server, in turn, wrap the generated HTML document in an HTTP response and sends it back to the browser.
- Most web applications store and retrieve data from a database that runs on the database server.
- The browser does not know or care whether the HTML was retrieved from a static HTML file or dynamically generated by the web application. Either way, the browser displays the HTML document that is returned.

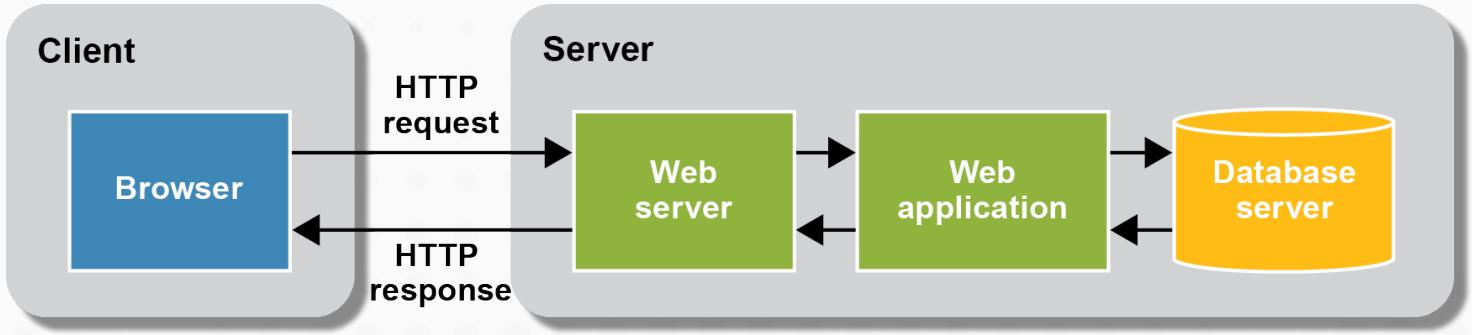
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STATIC & DYNAMIC WEB PAGES WORK

How a web server processes static web pages?



How a web server processes dynamic web pages?



THREE APPROACHES FOR DEVELOPING JAVA WEB APPLICATION

Servlet/JSP

- Is a lower-level API that does less work for the programmer.
- Provides a high degree of control over the HTML/CSS/JavaScript that's returned to the browser.

JSF

- Is a higher-level API that does more work for the programmer.
- Makes it more difficult to control the HTML/CSS/JavaScript that's returned to the browser.

Spring Framework

- Is a higher-level API that does more work for the programmer.
- Provides a high degree of control over the HTML/CSS/JavaScript that's returned to the browser.

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Jakarta Enterprise Edition (JEE)

TOPIC 1

JEE or Jakarta Enterprise Edition, previously known as Java Platform, Enterprise Edition (Java EE) is a Java platform by Oracle.

It offers a set of specifications, extending the features offered within Java Standard edition, SE, by including enterprise features for application development such as distributed computing, web services and Java microservices.

The core component of Java EE includes Enterprise Java Beans, EJBs, Java Server Pages, JSP, and Java Database Connectivity, JDBC.

Java EE applications run on microservices or application servers that handle all the fundamental things such as transactions, security, scalability, concurrency and component management.

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Spring Framework

- Spring is the application development framework for Java EE.
- It is an open-source Java platform that provides supports for developing robust and large-scale Java applications.
- It also offers tons of extensions that are used for building all sorts of large-scale applications on top of the Java EE platform.
- It is a lightweight framework that enables developers to develop enterprise-class applications using Plain Old Java Object, POJO.
- Although the Spring framework does not have any specific programming model, it has become popular among Java developers as an addition to the Java platform.
- It has a huge community of Java developers who are working and contributing to introducing more extensions and improving existing features offered by the Spring framework.

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JEE vs Spring

The basis of comparison	JEE	Spring
Architecture	Based on three-dimensional Architectural Framework, i.e. Logical Tiers, Client Tiers, and Presentation Tiers.	It is based on a layered architecture that includes many modules. These modules are made on top of their core container.
Language	It uses a high-level object-oriented language that has a certain style and syntax.	It doesn't have a certain programming model.
Interface	It typically has a graphical user interface created from Project Swing or Abstract Window Toolkit APIs.	Syntax the same everywhere – independent of an IDE or a compiler
Dependency Injection	Uses dependency injection	Uses dependency injection
Structure	Can be web-based or non-web-based	Based on almost 20 modules
Speed	Quite good speed	Spring is slower than Java EE

Reference: https://www.educba.com/java-ee-vs-spring/

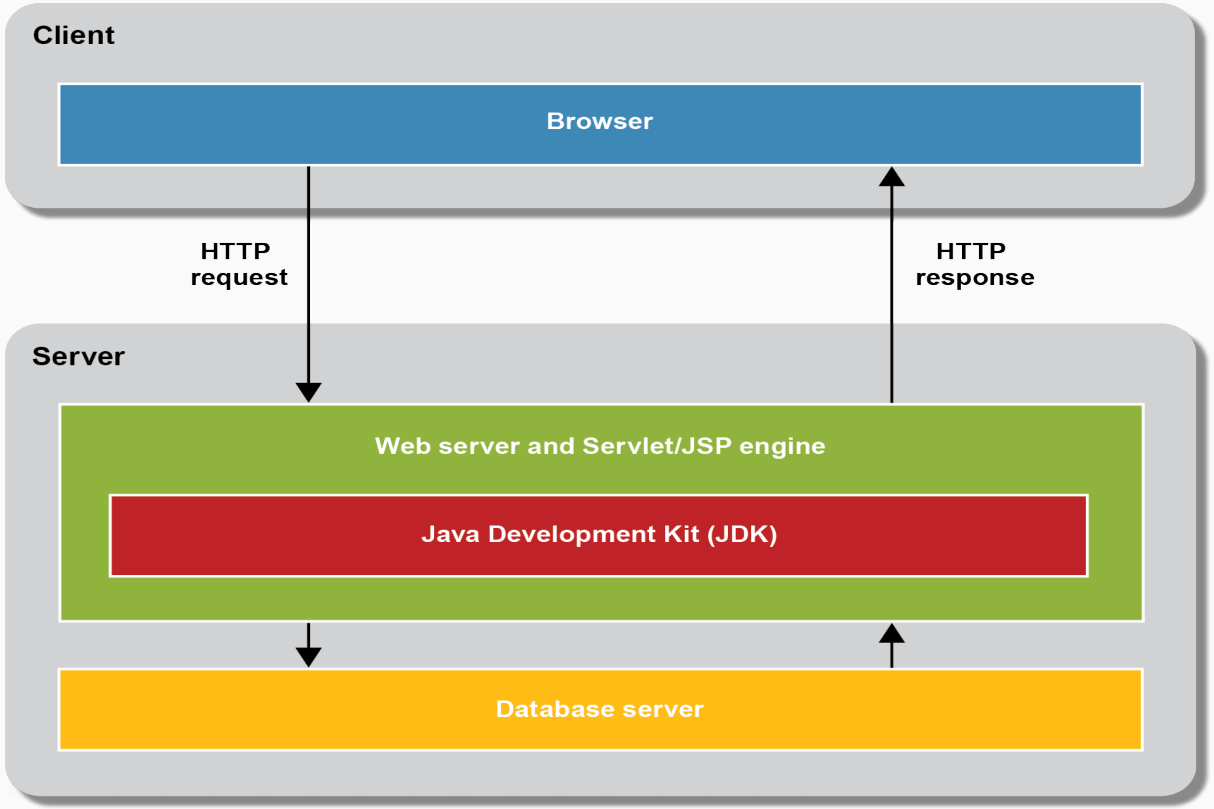
TERMINOLOGY

- The Java Standard Edition (Java SE) includes the Java Development Kit (JDK) and the Java Runtime Environment (JRE).
- The Java Enterprise Edition (Java EE) specification describes how web servers can interact with all Java web technologies.
- Servlets store the Java code that does the server-side processing.
- JavaServer Pages (JSPs) store the HTML that defines the user interface.
- JavaServer Faces (JSF) provides a higher-level API that replaces both servlets and JSPs.
- Java Persistence API (JPA) is an API for working with databases.

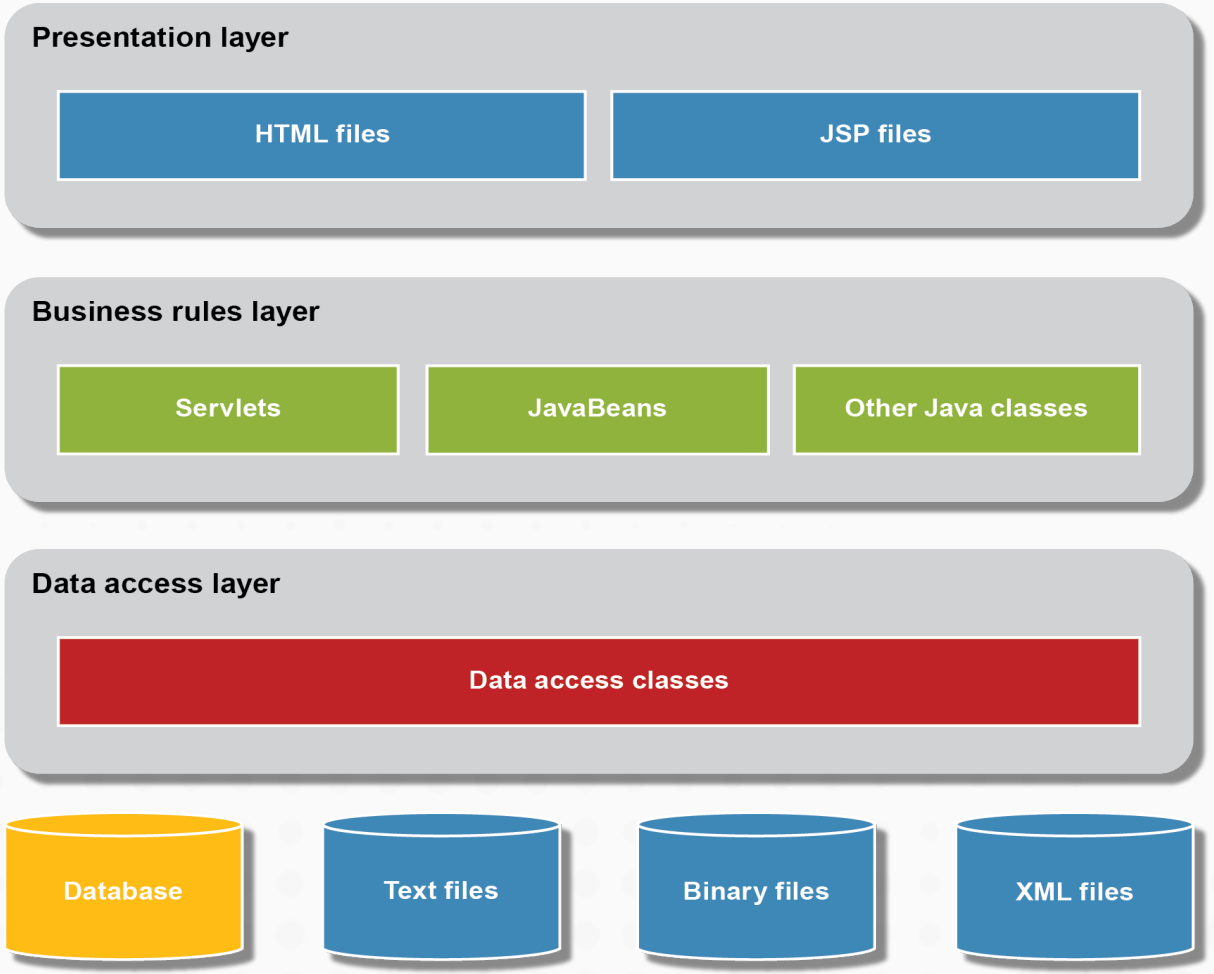
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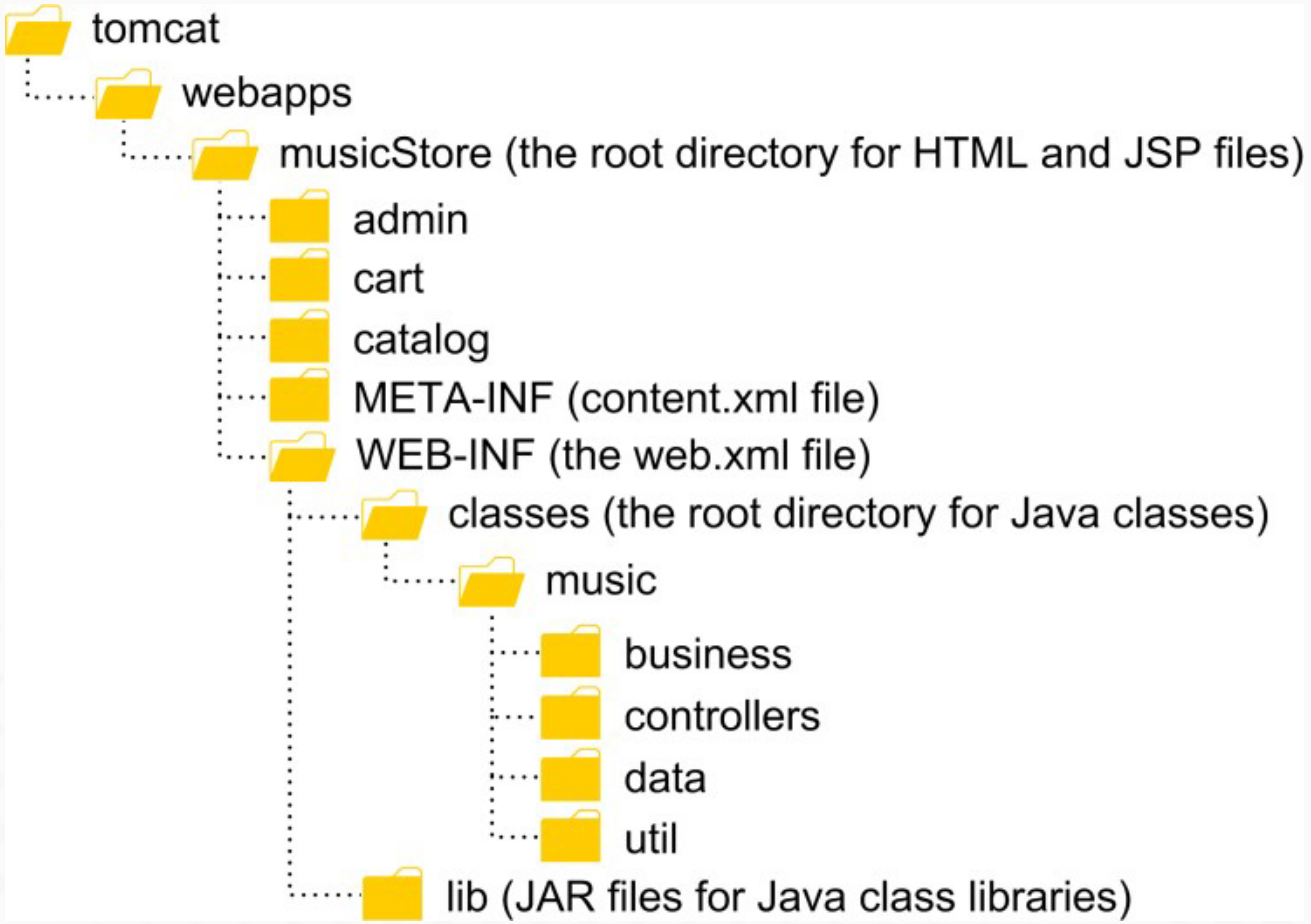
COMPONENTS OF SERVLET/JSP APPLICATION



ARCHITECTURE OF A TYPICAL SERVLET/JSP APPLICATION



THE DIRECTORY STRUCTURE FOR JAVA WEB APPLICATION



THE DIRECTORIES AND FILES FOR JAVA WEB APPLICATION

Directory	Description
(root)	Contains the HTML and JSP files.
\WEB-INF	Contains the web.xml file and is not directly accessible from the web.
\WEB-INF\classes	Contains the servlets and other Java classes for your application. Each subdirectory corresponds with the package for the Java class.
\WEB-INF\lib	Contains any JAR files that contain Java class libraries that are used by the web application.
\META-INF	Contains the context.xml file that configures the web application context.

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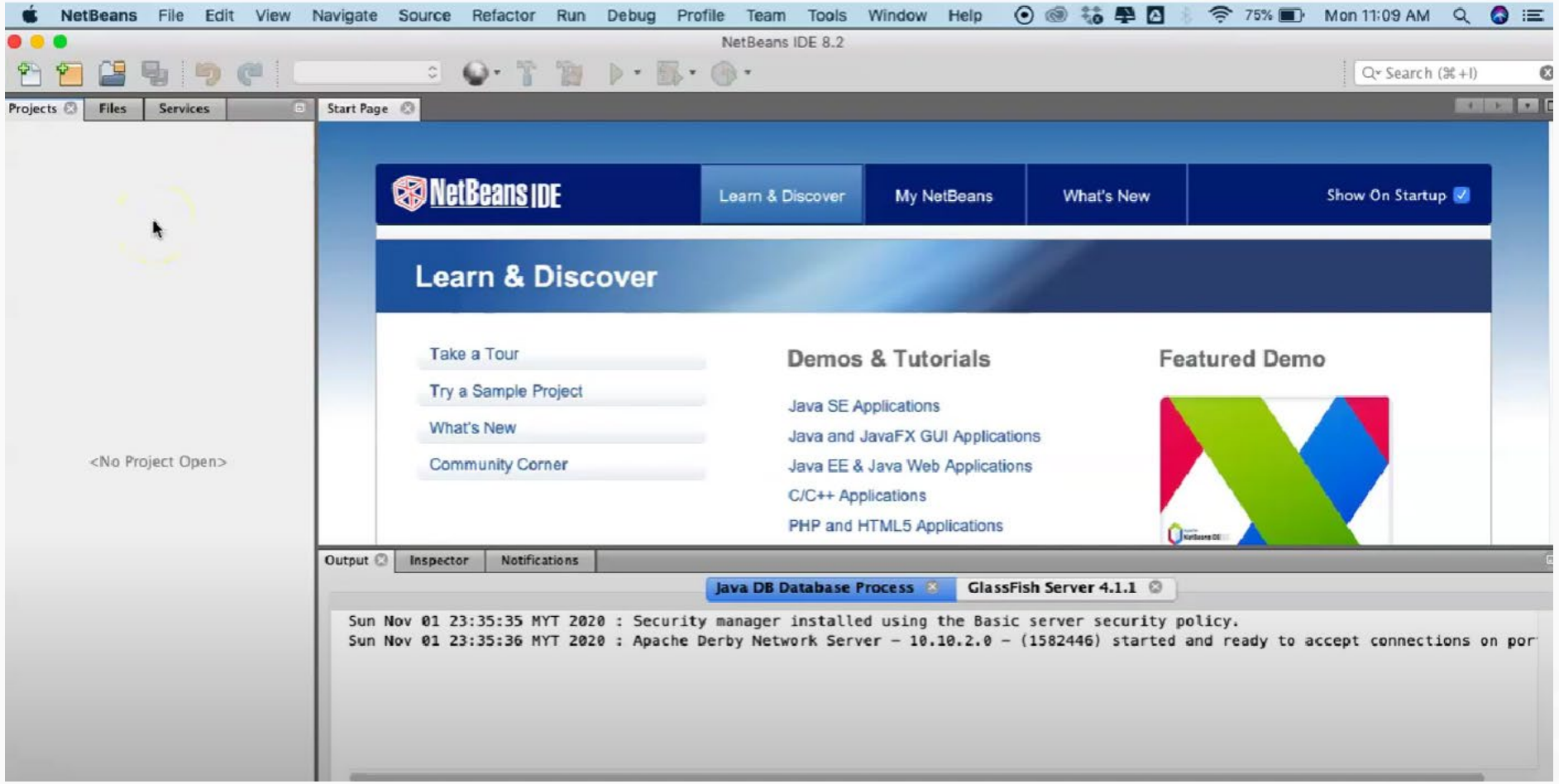
POPULAR IDEs FOR JAVA WEB DEVELOPMENT

- NetBeans
- Eclipse
- STS (Spring Tool Suite)
- Visual Studio Code + Extensions
- IntelliJ IDEA

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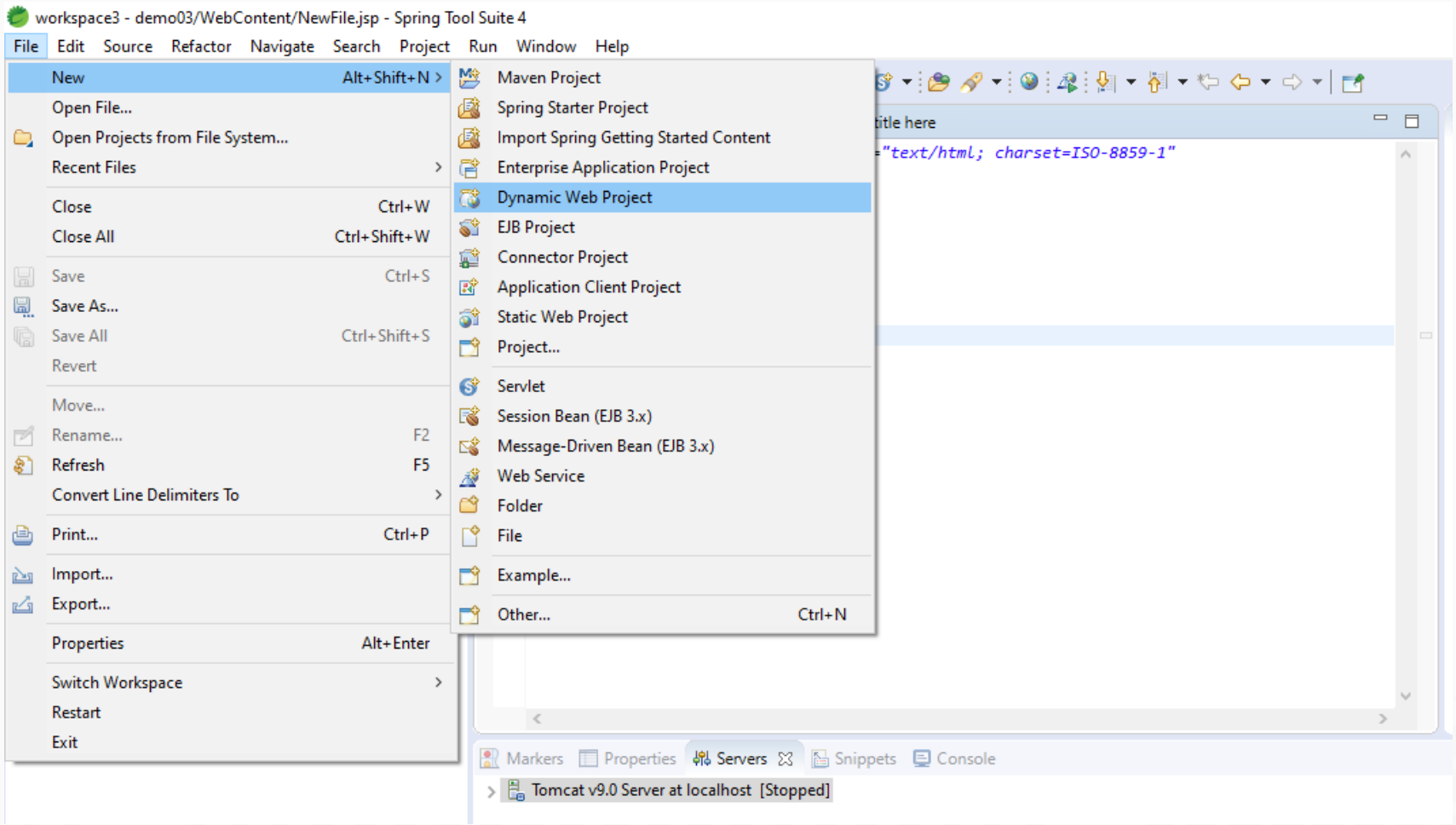
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NETBEAN IDE



TOPIC 1

SPRING TOOL SUITE (STS) IDE



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POPULAR WEB SERVERS



Tomcat

- Is a servlet/JSP engine that includes a web server.
- Is free, open-source, and runs on all modern operating systems.
- Is a popular web server for Java web applications.



GlassFish

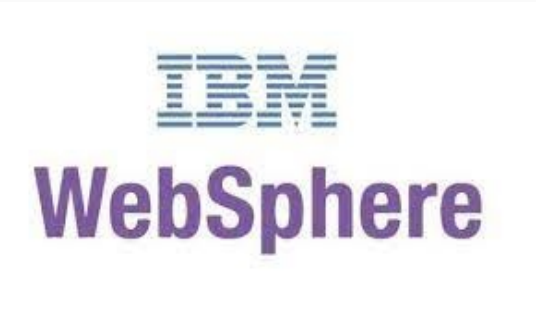
- Is a complete Java EE application server.
- Is free, open-source, and runs on all modern operating systems.
- Provides more features than Tomcat.
- Requires more system resources than Tomcat.

OTHER POPULAR WEB SERVERS

- WildFly (formerly JBoss)
- Jetty
- Oracle WebLogic
- IBM WebSphere









POPULAR DATABASE SERVERS

TOPIC 1

MySQL

- Is a relational database server that implements most of the SQL standard.
- Is free, open-source, and runs on all modern operating systems.
- Is the most popular database server for Java web applications.

Other popular database servers

- PostgreSQL
- Oracle



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Tools for deploying Java Web Application

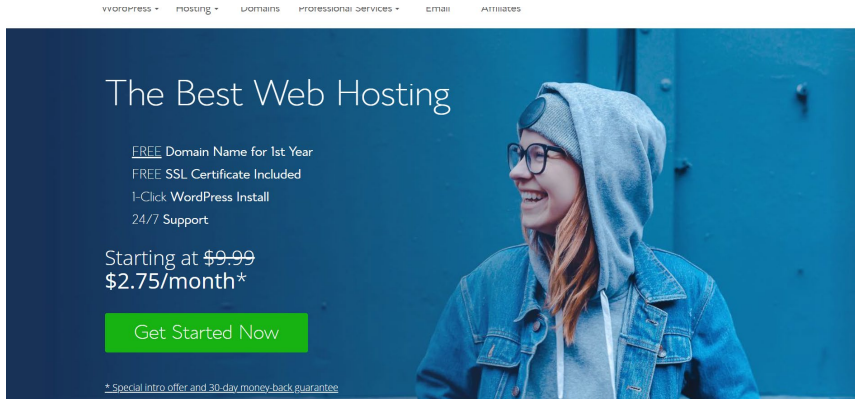
1-JSP Hosting

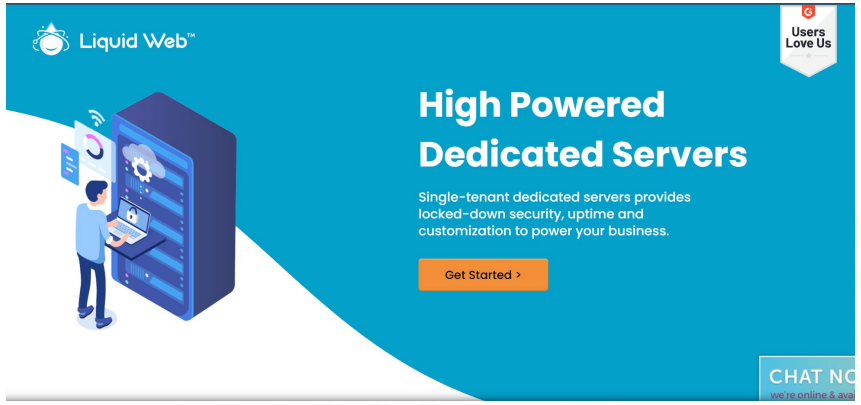
- A2 Hosting
- Liquid Web
- Bluehost

References:

www.a2hosting.com
www.liquidweb.com
www.bluehost.com

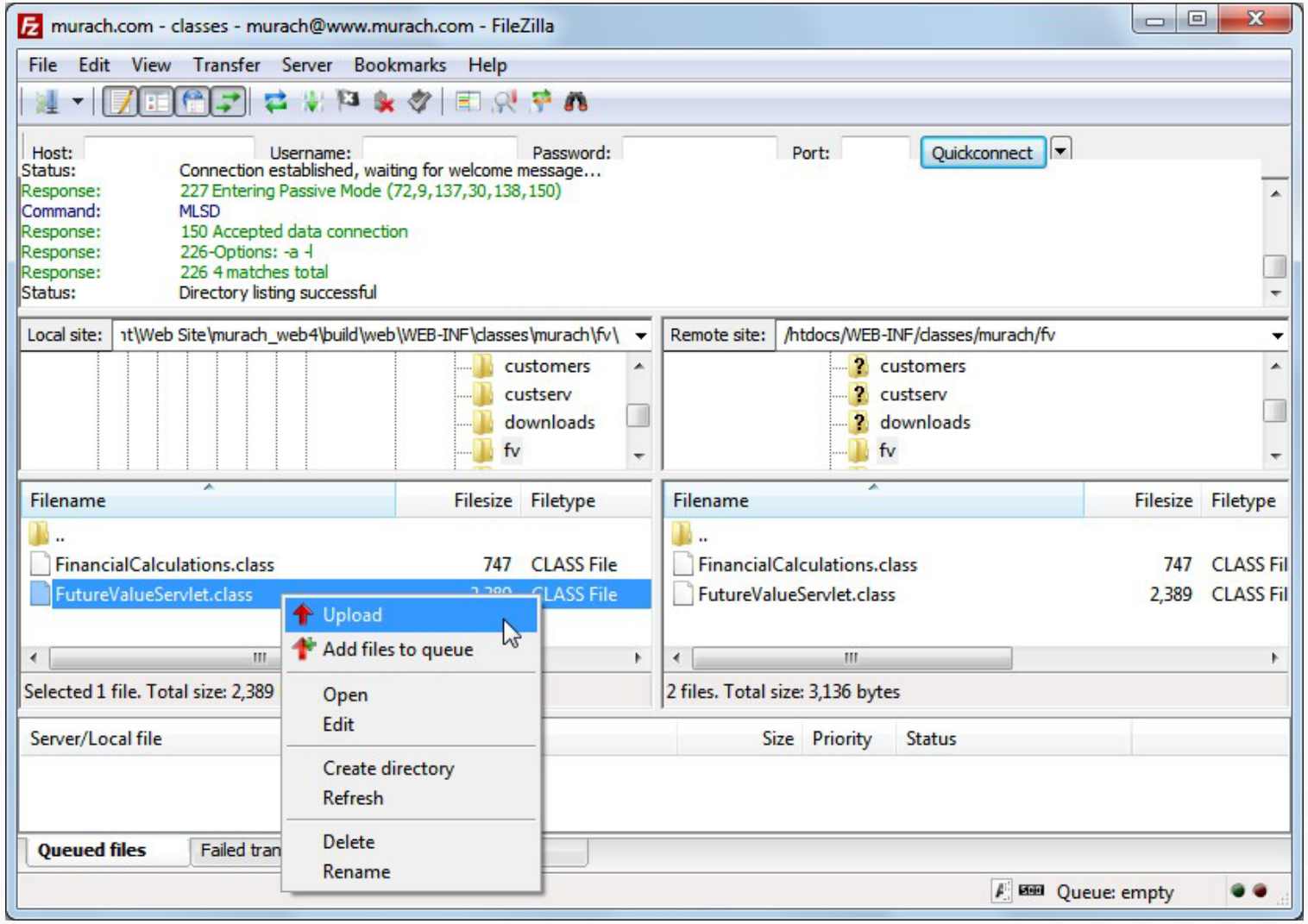






2- File Transfer Protocol (FTP): FILEZILLA

TOPIC 1



Reference: Murach's Java Servlets/JSP (3rd Ed.), 2014

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SUMMARY

TOPIC 1

- Understand the concepts and terms before starting web programming with Servlets and Java Server Pages (JSPs).
- Describe approaches that can use for developing Java web application.
- Describe software that can use to develop and deploy servlet/JSP application.

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